APPLICATIONS And ADVANTAGES Of Using Li-Fi For DATA TRANSFER

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Visible light communication (VLC) is a data communication method that uses visible light between 400THz and 800THz (Tera-Hertz) frequency; wavelength is 780nm to 375nm. It is a subset of optical wireless communication technologies.

VLC technology uses fluorescent lamps or light emitting diodes (LEDs) to transmit signals. Photosensitive components are used to receive those signals. However, in some cases a cellphone camera or a digital camera can also be used (since the camera is an array of photosensitive components arranged in a definite pattern). Such sensors may provide high input efficiency, since a large number of photosensitive components receive the signals.

VLC does not require a separate system to operate—it can be added to the illumination device. This article explores a new approach for VLC where LEDs are the transmitter, visible light is the medium for communication and a photodiode is the receiver. Thus, an attempt has been made to transmit data from LEDs to the photodiode.

Advancements in VLC lead to Light-Fidelity, or Li-Fi. Li-Fi is a bi-directional, high-speed and fully networked wireless communication technology, which is similar to Wi-Fi, or Wireless-Fidelity.

Outline of the operation of a VLC system is explained next, using block diagrams. Block diagram in Fig. 1 shows the process involving data transmission using visible light, or Li-Fi.

Transmission process block diagram. Block diagram in Fig. 2 shows the process of the transmission. Data is first obtained from the user and then converted into bits by a microcontroller (MCU) (Arduino is used in this case). Then, on-off keying (OOK) modulation technique is applied, and light pulses are generated from the light source.

Receiver process block diagram. Block diagram in Fig. 3 shows the process of the receiver operation. Light pulse obtained from the light source is sensed by photosensitive elements (LDR in this case). Sensed pulse is demodulated and bits are identified. Transmitted data is framed from received bits.

VLC transmitter

A transmitter module contains a power supply, LEDs and Arduino MCU. It converts electrical waves into optical pulses. VLC transmitter is different from a conventional communication transmitter. It acts as a communication transmitter and an illumination device simultaneously. For each character, seven bits are assigned uniquely. Each character is identified by a 7-bit combination. LEDs have a high flickering rate. Hence, these can be set to switch off and

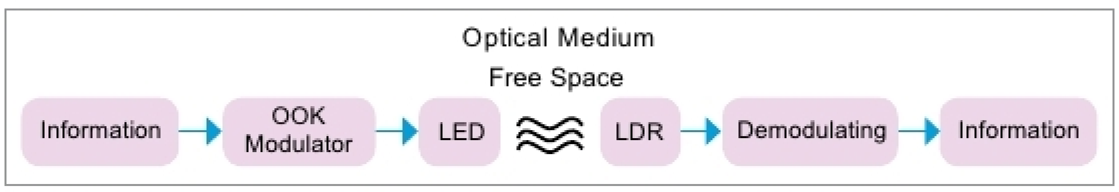

Fig. 1: VLC process block diagram

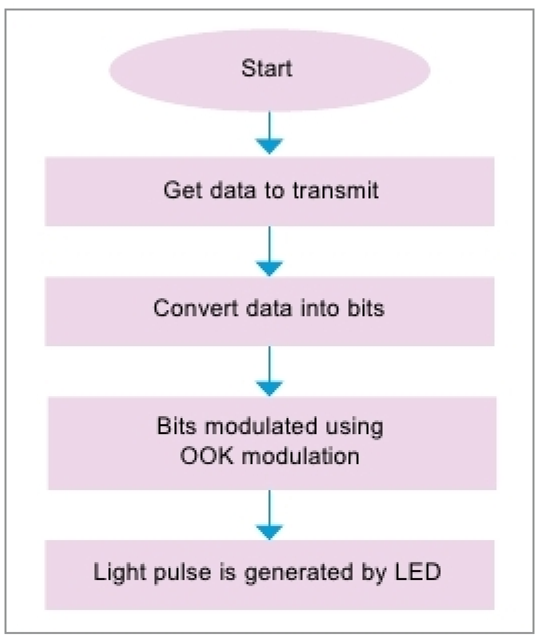

Fig. 2: Transmission process block diagram

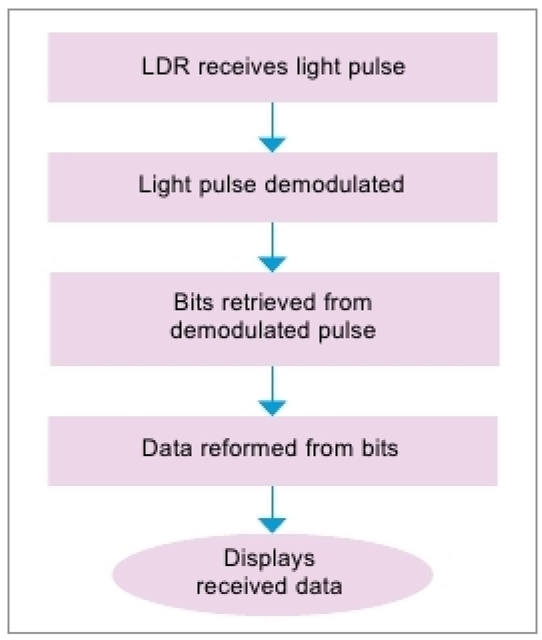

Fig. 3: Receiver process block diagram

Li-Fi, which uses visible light for communication, is 100 times faster than Wi-Fi, which uses radio frequencies